

Fig. 5: An actual-size PCB layout for the power supply and microcontroller unit

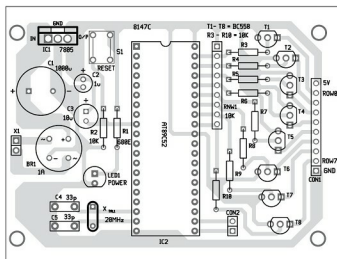


Fig. 6: Component layout of power supply and microcontroller unit

provides the 5V output voltage for driving the circuit around microcontroller IC2. AT89C52 (IC2) is a low-power, high-performance CMOS 8-bit microcontroller. It has 8k bytes of Flash ROM, 256 bytes of RAM, 32 I/O lines, three 16-bit timers/counters, a six-vector two-level interrupt, a full duplex serial port, an on-chip oscillator

and on-clock circuitry.

CD4094 (IC3-IC5) is an 8-stage serial shift register, having a storage latch associated with each stage for stroking data from the serial input to parallel buffered 3-state outputs. The parallel outputs may be connected directly to common bus lines. Data is shifted on positive clock transition. Data in each

shift register stage is transferred to the storage register when the strobe input is high. Data in the storage register appears at the outputs, whenever the Output-Enable signal is high. Two serial outputs are available for cascading a number of CD4094 devices. Data is available at the Q serial output terminals on positive clock edges to allow

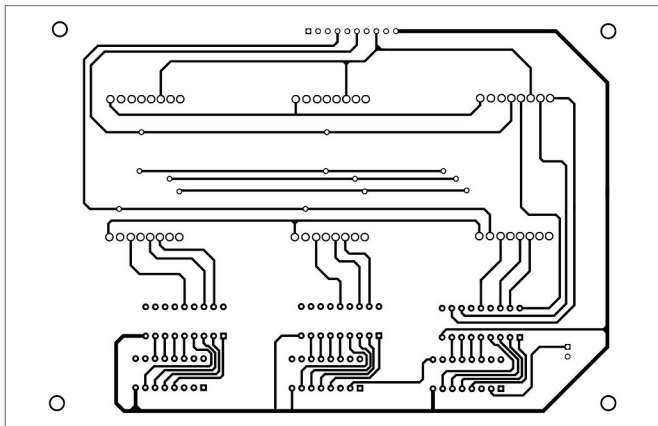


Fig. 7: Track layout of the bottom layer of display unit